

```
1 from flask import Flask, jsonify, request
2 app = Flask(__name__)
3 BOOKS = [
4     {"id": "abc-1234", "t": "Python Programming", "author": "Guido"},
5     {"t": "Flask Web Development", "author": "Miguel"},
6     {"t": "Clean Code", "author": "R. Martin"}
7 ]
8 def find_by_id(book_id):
9     return next((b for b in BOOKS if b.get("id") == book_id), None)
10
11 # Path params: /books/<id> (id là string)
12 @app.route("/books/<book_id>", methods=["GET"])
13 def get_book(book_id):
14     book = find_by_id(book_id)
15     if book is None:
16         return jsonify({"error": "not found"}), 404
17     return jsonify(book), 200
18 @app.route("/items/<int:item_id>")
19 def get_item(item_id): # int sẵn
20     return jsonify({"id": item_id}), 200
21 @app.route("/books", methods=["GET"])
22 def list_books():
23     limit = int(request.args.get("limit", 20))
24     q = request.args.get("q", "").strip().lower()
25     items = [b for b in BOOKS if q in b["t"].lower()]
26     return jsonify({"items": items[:limit]}), 200
27 if __name__ == "__main__":
28     app.run(host="127.0.0.1", port=5000, debug=True)
```

```
THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA/api_session_01
$ curl -i "http://127.0.0.1:5000/books?q=python&limit=5"
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 11 Sep 2026 13:42:30 GMT
Content-Type: application/json
Content-Length: 114
Connection: close

{
  "items": [
    {
      "author": "Guido",
      "id": "abc-1234",
      "t": "Python Programming"
    }
  ]
}
```

```
THI@LAPTOP-HCT3QG47 MINGW64 /e/SOA/api_session_01
$ curl -i http://127.0.0.1:5000/books/abc-1234
HTTP/1.1 200 OK
Server: Werkzeug/3.1.8 Python/3.12.3
Date: Fri, 11 Sep 2026 13:42:55 GMT
Content-Type: application/json
Content-Length: 73
Connection: close

{
  "author": "Guido",
  "id": "abc-1234",
  "t": "Python Programming"
}
```

